

Stage 4 Boundary Diagnostic Report

Premium-Implied Boundaries and a Direct Boundary-Head Baseline

Codex-assisted surrogate modelling review

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Abstract

This report reviews Stage 4 of the compressed roadmap. It checks whether the Stage 3 positive-premium surrogate can recover early-exercise boundary behavior from predicted continuation premium, and compares it with a small direct boundary-head baseline. Boundary labels remain threshold-based CN/PSOR diagnostics, not exact analytical free boundaries. No Delta or Gamma head is trained. The Stage 4 review decision is `READY_FOR_DELTA_DIAGNOSTIC_STAGE`.

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1 Purpose of Stage 4

Stage 4 is a diagnostic boundary experiment. Stage 3 showed that predicting continuation premium helped preserve the American payoff obstacle. The next question is whether that premium surrogate also gives coherent early-exercise boundary behavior when the same threshold extraction logic is applied to predicted premium curves. The boundary interpretation follows the American-option finite-difference and free-boundary setting documented in the solver reports (Brennan and Schwartz, 1977; Wilmott et al., 1995).

2 Link to Stage 3 Result

The Stage 3 review decision was `READY_FOR_BOUNDARY_DIAGNOSTIC_STAGE`. Stage 4 uses the same v1 small-grid dataset and retrains the positive-premium MLP deterministically because Stage 3 did not save model weights.

3 Boundary Target Construction

The v1 dataset stores `boundary_spot_over_K` on sampled rows. Stage 4 collapses those rows to one target per regime-time pair. The resulting target table contains 34,848 rows, with 29,795 finite reference boundaries and 20,093 finite boundaries inside the sampled $S/K \in [0.4, 1.8]$ support. There are 4,356 no-dividend call control time rows.

4 Premium-Implied Boundary Method

The premium-implied method retrains the Stage 3 positive-premium MLP in memory, predicts `premium_over_K` over v1 sampled rows, sorts each regime-time curve by moneyness, and applies the Ticket 09 threshold crossing with threshold 10^{-6} . It is a diagnostic extraction, not a new solver. In this Stage 4 test, the premium-implied boundary method is not reliable: it often has no comparable boundary-error points, so its RMSE is reported as `NA` rather than zero error.

5 Direct Boundary-Head Method

The direct boundary head is a small baseline. It uses only `tau_fraction`, r , q , σ , T , and `is_call`. A histogram-gradient classifier predicts whether a boundary exists, and a bounded histogram-gradient regressor predicts `boundary_spot_over_K` for finite-boundary rows. Maturity rows and no-dividend call controls are postprocessed as no-boundary diagnostic cases. No Delta, Gamma, or production boundary head is introduced.

6 Metrics and Evaluation Protocol

Metrics include boundary MAE, RMSE, maximum absolute error, found/not-found agreement, missed boundary rate, false boundary rate, and no-dividend-call false-boundary rate. Boundary errors are computed only where a reference boundary exists, the prediction found a boundary, and the reference boundary is inside the sampled support.

7 Results by Split

Method	Split	RMSE	Found agreement	No-div false rate
premium_implied_boundary	train	NA	0.022911	NA
premium_implied_boundary	validation	NA	0.008264	NA
premium_implied_boundary	test	NA	0.855276	0
premium_implied_boundary	stress_holdout	NA	0.008264	NA
direct_boundary_head	train	0.030943	0.999918	NA
direct_boundary_head	validation	0.035362	1.000000	NA
direct_boundary_head	test	0.041007	0.993273	0
direct_boundary_head	stress_holdout	0.084293	1.000000	NA

Stage 4 passes because the direct boundary-head baseline satisfies the boundary diagnostics, not because the premium-implied threshold extraction succeeds. The premium-implied method is useful as a diagnostic stress test of the price/premium surrogate, but it did not recover dependable boundary curves in this run. This supports using a separate boundary-focused model for boundary diagnostics instead of relying only on price/premium threshold extraction.

8 Results by Option Family

Method	Option	RMSE	MAE	Found agreement
premium_implied_boundary	put	NA	NA	0.008264
premium_implied_boundary	call	NA	NA	0.281738
direct_boundary_head	put	0.032305	0.023906	1.000000
direct_boundary_head	call	0.046284	0.031983	0.997876

9 Sample Boundary Curves

10 Limitations

Boundary labels are threshold-based diagnostics from the finite-difference solver. They are not analytical free-boundary truth. Premium-implied extraction is limited by the sampled S/K window, and some reference boundaries lie outside that support. The direct boundary head is a small diagnostic baseline, not a solver replacement.

11 What Stage 4 Supports

Stage 4 supports comparing premium-implied and direct boundary diagnostic behavior by split, option family, and sampled boundary curves. It also records no-dividend call false-boundary behavior as a control. The evidence supports a separate boundary-focused model as the coherent Stage 4 boundary diagnostic path.

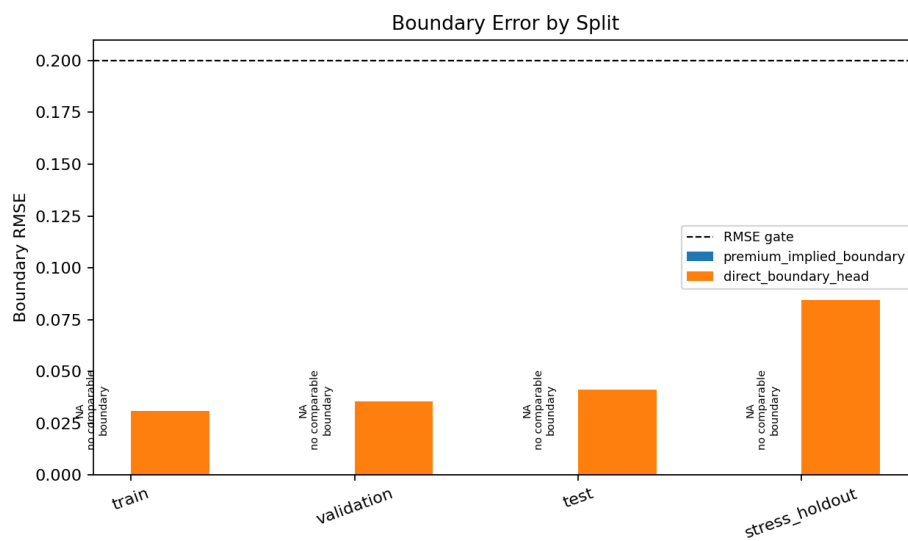


Figure 1: Boundary error by split.

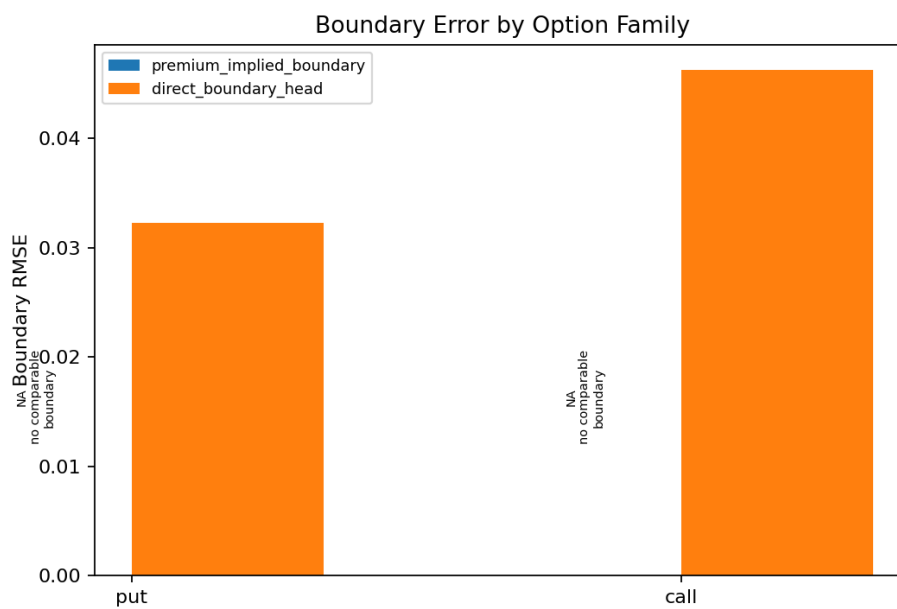


Figure 2: Boundary error by option type.

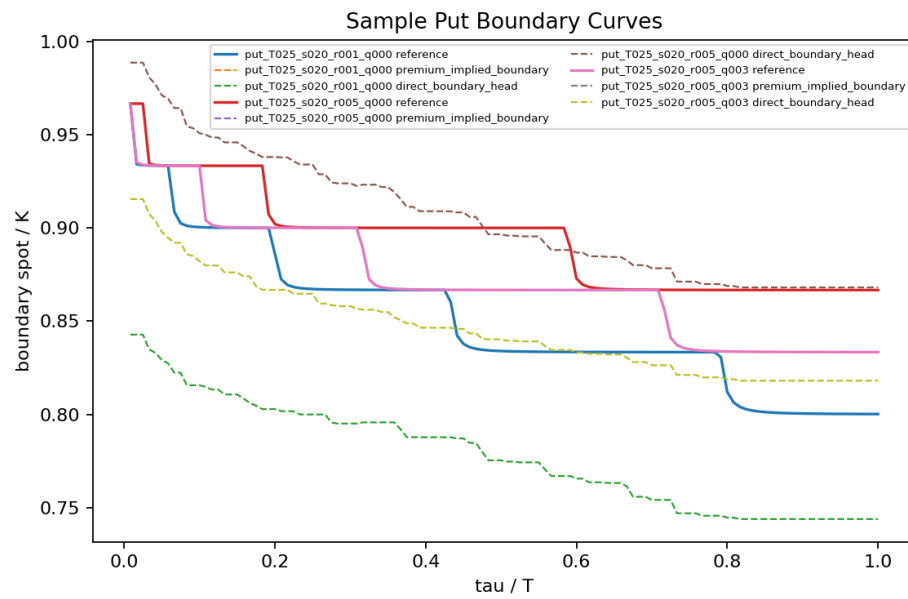


Figure 3: Sample put boundary curves.

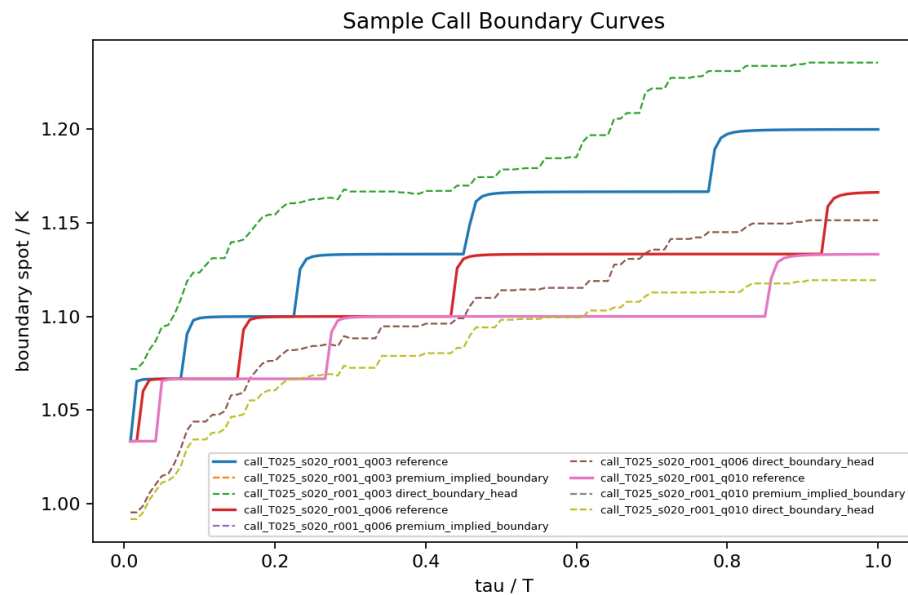


Figure 4: Sample call boundary curves.

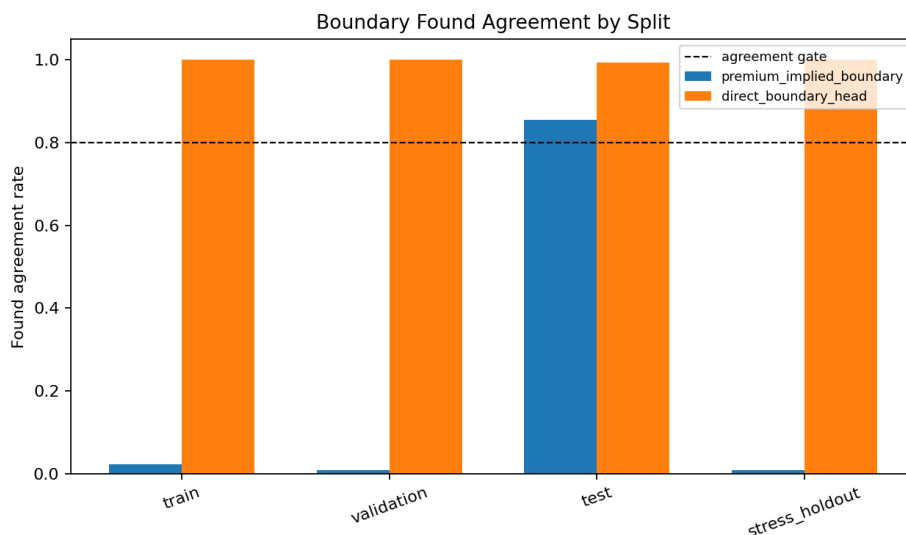


Figure 5: Boundary found rate comparison.

12 What Stage 4 Does Not Support

This stage does not support production boundary reliability, Delta or Gamma target training, exact free-boundary claims, broad architecture search, or final surrogate conclusions.

13 Recommended Next Stage

If the decision is `READY_FOR_DELTA_DIAGNOSTIC_STAGE`, the next stage may plan a cautious Delta diagnostic experiment. If the decision is `REVIEW_REQUIRED_BEFORE_DELTA_STAGE`, human review should inspect boundary failures before adding any new diagnostic heads.

References

- Michael J. Brennan and Eduardo S. Schwartz. The valuation of american put options. *The Journal of Finance*, 32(2):449, 1977. doi: 10.2307/2326779.
- Paul Wilmott, Sam Howison, and Jeff Dewynne. *The Mathematics of Financial Derivatives*. Cambridge University Press, 1995. doi: 10.1017/CBO9780511812545.